

# CS 486/686

# Decision Trees

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Lecture 16

RN 19.3, 19.8 · PM 7.3.1, 7.5

# Reminder: next week is asynchronous



I'm away at ICML next week, so **L17 (Tue Jul 7)** and **L18 (Thu Jul 9)** are **pre-recorded videos** — **no in-person class** those days. Watch on your own time (links posted before each class).



**Deadlines are unchanged:** Chat 8 + the CS686 project proposal are due **Tue Jul 7**; Chat 9 is out and **Assignment 2 is due Thu Jul 9**.



Questions? Post on **Piazza** — the TAs are available all week, and I'll follow up when I'm back.

Search ›

Uncertainty ›

Decisions ›

Learning

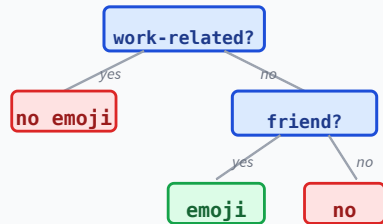
## Learning goals

- Describe the **components** of a decision tree.
- **Construct** a tree from a feature-testing order.
- Compute a tree's **test-set accuracy**.
- Compute **entropy** and **information gain**; trace the **DT learner**.
- Use **Gini impurity**; control overfitting with **pruning**.
- Combine trees: **random forests** and **gradient boosting**.

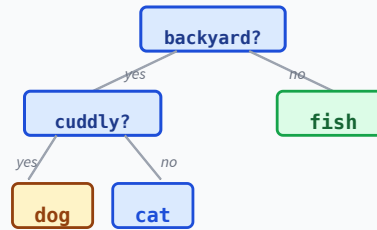
# Decision trees in the wild

A decision tree is nothing more than a **nested if-then-else** — learned automatically from data.

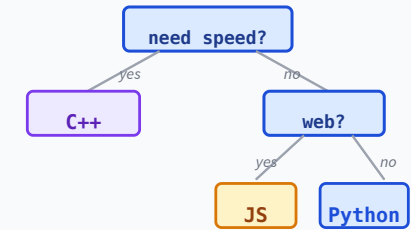
Should I use an emoji?



What pet?



Which language?



Simple to read, fast to evaluate, and surprisingly effective in practice.

# Running example: Jeeves the valet

**Predict from the weather:** will Bertie play tennis today? Jeeves has logged 14 mornings.

## Features and target

- **Features:** Outlook (Sunny/Overcast/Rain), Temp (Hot/Mild/Cool), Humidity (High/Normal), Wind (Weak/Strong).
- **Target:** Tennis? (Yes / No).

Learn from the training days → evaluate on a held-out test set.

# The training set (14 days)

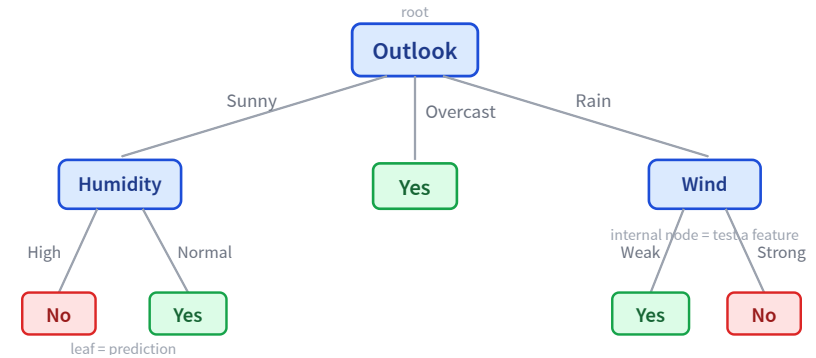
| Day | Outlook  | Temp | Humidity | Wind   | Tennis? |
|-----|----------|------|----------|--------|---------|
| 1   | Sunny    | Hot  | High     | Weak   | No      |
| 2   | Sunny    | Hot  | High     | Strong | No      |
| 3   | Overcast | Hot  | High     | Weak   | Yes     |
| 4   | Rain     | Mild | High     | Weak   | Yes     |
| 5   | Rain     | Cool | Normal   | Weak   | Yes     |
| 6   | Rain     | Cool | Normal   | Strong | No      |
| 7   | Overcast | Cool | Normal   | Strong | Yes     |

| Day | Outlook  | Temp | Humidity | Wind   | Tennis? |
|-----|----------|------|----------|--------|---------|
| 8   | Sunny    | Mild | High     | Weak   | No      |
| 9   | Sunny    | Cool | Normal   | Weak   | Yes     |
| 10  | Rain     | Mild | Normal   | Weak   | Yes     |
| 11  | Sunny    | Mild | Normal   | Strong | Yes     |
| 12  | Overcast | Mild | High     | Strong | Yes     |
| 13  | Overcast | Hot  | Normal   | Weak   | Yes     |
| 14  | Rain     | Mild | High     | Strong | No      |

9 Yes, 5 No. We'll learn a tree from these 14 rows.

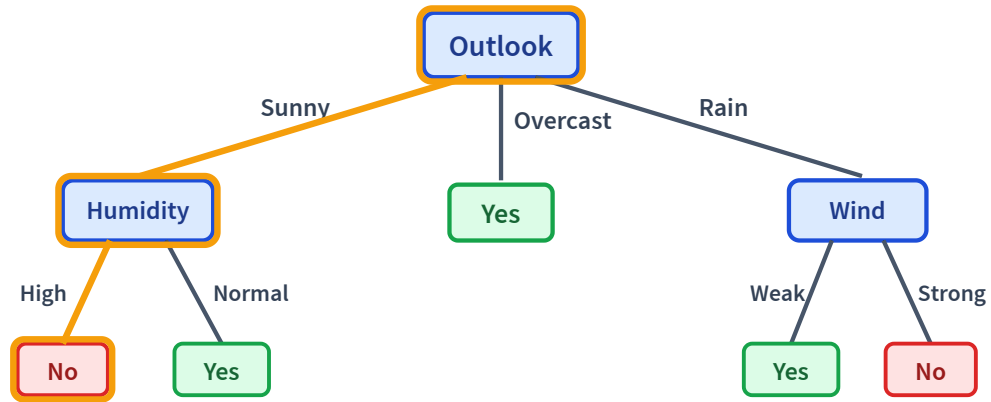
# Anatomy of a decision tree

- A supervised classifier with a **single discrete target**.
- Each **internal node** tests an input feature.
- **Edges** are labeled with feature values.
- Each **leaf** predicts a target class.
- **Classify**: follow edges from the root down to a leaf.



Edges are labeled with feature values (e.g. Sunny / Overcast / Rain).

# Classify an example: walk down the tree



Follow the tests from the root down to a leaf.

**New day:** Outlook = Sunny,  
Humidity = High.

Root tests **Outlook** → Sunny.

Then test **Humidity** → High.

Reach a leaf → predict **No**.

The path is just a nested if-then-else  
— easy to turn into code.



# Build a tree given an order

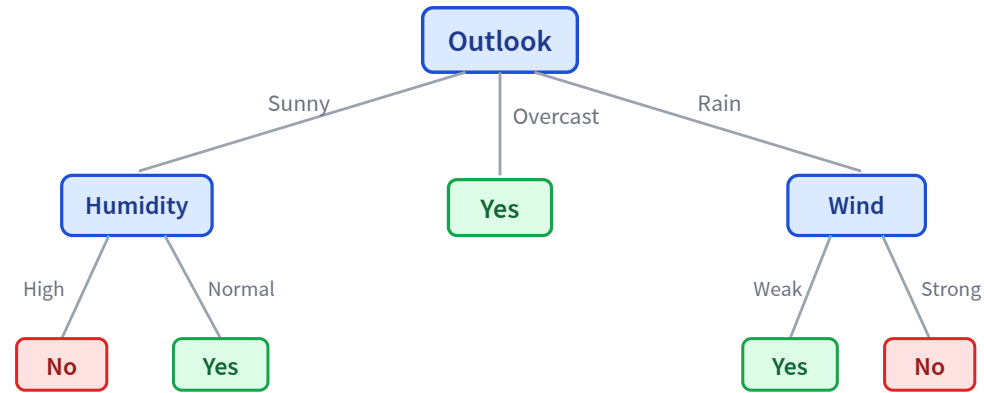
Pick a feature-testing order; recursively split the training examples by that feature's value.

## An order for the Jeeves data

- **Root:** test Outlook.
- Sunny → test **Humidity**.
- Overcast → all Yes, done.
- Rain → test **Wind**.

Each example flows down the tree according to its feature values; leaves end up with subsets of the training set.

# The Jeeves decision tree



Built from the 14 training rows by following the feature order above.

# When do we stop? Three cases

## All same class

Every example at this node has the same label.

Return that label.

## No features left

We've tested every feature, but examples still disagree (noisy data).

Return the **majority class** at this node.

## No examples left

A feature value unseen in training — no examples reach here.

Return the **parent's majority class**.

"No features left" → noisy labels; "no examples left" → unseen feature combinations.

# The decision-tree learner

## DT-Learner(*examples*, *features*)

1. If all examples are in the same class, return that class.
2. Else if no features left, return the majority class of *examples*.
3. Else if no examples left, return the majority class at the *parent*.
4. Else: choose a feature  $f$ ; for each value  $v$  of  $f$ ,
  - build an edge labeled  $v$ ;
  - recurse on examples with  $f = v$  and features  $\setminus \{f\}$ .

The only open question: **which feature do we pick to test?**

# Which feature should we test?

Pick the feature whose split makes the children most **class-pure**.

## Split on Outlook



Overcast cleanly predicts **Yes** — one branch is pure!

## Split on Temp



Every Temp value still mixes Yes/No — messier.

We need a metric to score "purity". Enter **entropy**.

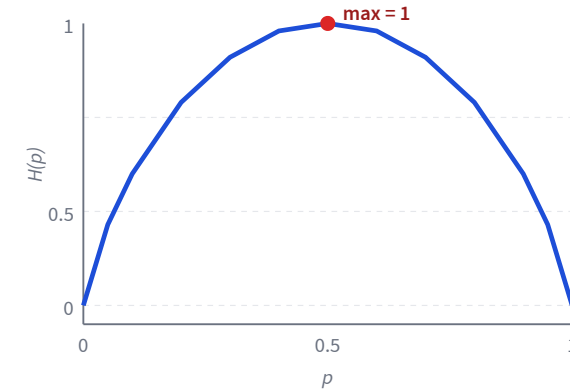
# Entropy: measuring uncertainty

## Entropy

$$I(P(c_1), \dots, P(c_k)) = - \sum_{i=1}^k P(c_i) \log_2 P(c_i)$$

Bits of uncertainty. Bigger = less certain.

- $I(0.5, 0.5) = 1$ : maximally uncertain.
- $I(0.01, 0.99) \approx 0.08$ : nearly certain the second outcome.
- $I(1, 0) = I(0, 1) = 0$ : no uncertainty at all.



# Expected information gain

Split on a feature with  $k$  values  $\rightarrow$  compare the entropy *before* vs. the weighted entropy *after*:

$$\text{before (parent): } H_{\text{before}} = I\left(\frac{p}{p+n}, \frac{n}{p+n}\right)$$

$$\text{after (children): } H_{\text{after}} = \sum_{i=1}^k \frac{p_i + n_i}{p + n} I\left(\frac{p_i}{p_i + n_i}, \frac{n_i}{p_i + n_i}\right)$$

$$\text{gain: } IG = H_{\text{before}} - H_{\text{after}}$$

Pick the feature with the **largest** information gain at every node.

# Jeeves: gain at the root (Outlook)

Training set has 9 Yes, 5 No ( $p = 9, n = 5$ ).

**Q3.**  $H_{\text{before}} = I(9/14, 5/14) = ?$   
 $\Rightarrow \mathbf{0.940 \text{ bits.}}$

**Q4.** Split on **Outlook**: Sunny (2+/3-), Overcast (4+/0-), Rain (3+/2-).

$$H_{\text{after}} = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.694.$$
$$\Rightarrow IG(\text{Outlook}) = 0.940 - 0.694 = \mathbf{0.247}.$$



# Jeeves: gain at the root (Humidity) and winner

Same  $H_{\text{before}} = 0.940$ . Now try splitting on Humidity instead.

**Q5.** Split on **Humidity**: High (3+/4-), Normal (6+/1-).

$$H_{\text{after}} = \frac{7}{14}(0.985) + \frac{7}{14}(0.591) = 0.788.$$

$$\Rightarrow IG(\text{Humidity}) = 0.940 - 0.788 = \mathbf{0.151}.$$

**Q6.** Which feature do we pick as the root?

**Outlook** – higher IG:  $0.247 > 0.151$ .

# ID3: the canonical algorithm

## **ID3(examples, features)**

1. Apply the three base cases from DT-Learner (same-class / no-features / no-examples).
2. For each feature  $f$  in features, compute  $IG(f)$  on the current examples.
3. Choose  $f^* = \arg \max_f IG(f)$ ; make an internal node testing  $f^*$ .
4. For each value  $v$  of  $f^*$ : recurse on examples with  $f^* = v$  and features  $\setminus \{f^*\}$ .

Greedy: pick the locally best feature at each step. Not always globally optimal — but fast and effective.

# Another purity score: Gini

Information gain uses entropy. A cheaper alternative: how often we'd **mislabel** an example drawn at random from the node.

## Gini impurity

$$G = 1 - \sum_{i=1}^k P(c_i)^2$$

0 when the node is pure; maximal when classes are balanced — same spirit as entropy, but cheaper to compute (no logs).

Entropy vs. Gini rarely changes the tree much — both reward pure children.

# Trees overfit — so we prune

Grown to purity, a tree memorizes the training set (one leaf per noisy example) and generalizes poorly — a first taste of **overfitting** (we'll formalize it next lecture).

## Pre-pruning

Stop growing early: cap **max depth**, require a **minimum #examples** per node, or a minimum gain to split.

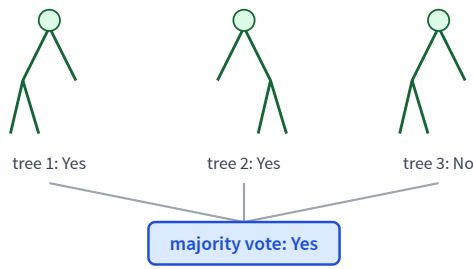
## Post-pruning

Grow the full tree, then **collapse** branches that don't improve accuracy on held-out data.

A smaller tree often generalizes **better** than one grown to purity.

# Random forests: a crowd of trees

One deep tree is high-variance. **Bagging** trains many trees, each on a **bootstrap sample** (draw  $n$  rows at random *with replacement* — some repeat, some are left out), then averages them; a **random forest** also splits on a random feature subset to decorrelate them.



- **Classification:** majority vote.
- **Regression:** average the predictions.
- Averaging cancels individual trees' errors → **lower variance**, strong off-the-shelf accuracy.

# Gradient boosting: fix your own mistakes

Instead of averaging independent trees, **add trees in sequence**. Let  $F_m(x)$  be the ensemble's prediction on input  $x$  after  $m$  trees (start with  $F_0 = 0$ ); each new tree corrects the errors made so far:

$$F_m(x) = F_{m-1}(x) + \gamma h_m(x)$$

- $h_m$  = the  $m$ -th small tree, fit to the current **residuals** (for squared loss, residual = true label —  $F_{m-1}(x)$ , the part not yet explained).
- $\gamma$  = a small learning rate — many shallow trees, each added slowly.
- **XGBoost** / **LightGBM** are the go-to winners on **tabular** data today.

Forests reduce *variance* (parallel); boosting reduces *bias* (sequential).

# Heads-up: I'm away next week

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# What's next: a refreshed ML/DL segment

I'm restructuring the machine learning part of the course to bring it up to date with where the field is today:



Stronger **foundations** (supervised learning, generalization) and modernized **classical methods**.



A proper **deep learning** track: neural nets → **Transformers & attention** → LLMs → **generative AI**.



Modern topics with real **mathematical depth**, not just buzzwords.

Slides may keep evolving as we go — **stay tuned!**



# Learning goals (recap) — Next: supervised learning

- ✓ Describe the **components** of a decision tree.
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L17: the general recipe behind supervised learning — **regression, loss functions, and generalization.**